Polar Format-Based Compressive SAR Image Reconstruction With Integrated Autofocus

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***Abstract*— In this article, we develop a polar format-based algorithm for recovering synthetic aperture radar (SAR) imagery from sparsely/randomly sampled data in the presence of motion compensation/phase errors. Our method integrates conventional SAR autofocus algorithms into an ℓ_1 norm penalized image recovery problem. We demonstrate the effectiveness of this approach using a real airborne SAR data set covering a large scene.**

***Index Terms*— Autofocus, compressed sensing, compressive sensing, sparse signal processing, synthetic aperture radar (SAR).**

I. Introduction

In RECENT years, the tremendous interest in the exploitation of sparsity in all domains of signal processing has been driven by strong provable guarantees on sparse recovery from the field of compressed sensing, combined with good empirical results even when provable recovery is not guaranteed. The field of radar signal processing has been no exception to the strong interest in compressed sensing/sparse signal-processing methods.

Synthetic aperture radar (SAR) imaging is a radar sensing modality whereby airborne or space-based radar systems can generate a high-resolution image by synthesizing a long synthetic antenna aperture using the motion of the platform. Many interesting results concerning the recovery of SAR imagery from sparsely sampled data have been demonstrated in the literature. We refer the reader to [1]–[6] as representative of prior work in the field.

Although interest in sparse recovery for SAR imaging has been strong, the practical motivation behind it for radar is in general rather less clear than for some other areas. For example, in medical imaging, being able to recover MRI images from a sparse sampling of wavenumber space has clear practical benefits as it reduces the time required to collect images and potentially reduces the barrier to use in certain clinical settings [7], [8]. Furthermore, such sparse sampling is physically practical in the context of MRI. In contrast, the same improvement is generally not possible with SAR, as an aircraft or satellite cannot randomly move across the synthetic aperture. However, in some cases, the ability to reconstruct SAR imagery from a random sampling of the nominally constant pulse-repetition frequency (PRF) synthetic aperture data may be useful. One case in which such random sampling may be useful is a radar system interleaving multiple modes such as moving target indication (MTI) and SAR or possibly interleaving radar and communications functions. One can imagine a radar system attempting to maintain a track on a ground target or airborne target and also needing to image some scene on the ground. In such cases, it may be undesirable to completely cease MTI operation while a SAR image is collected, which could take tens of seconds and may result in losing the track on an important target. One option would be to quasi-randomly interleave MTI pulses with SAR pulses. A similar situation may occur with a radar system that is supporting communications functions that must maintain a communications link while simultaneously collecting a SAR image. Both of these concepts would imply a random sampling in the slow-time or pulse domain. An alternative situation in which random sampling of the radar data can occur is collecting ultrawideband data while other users (communication systems or radars) occupy part of the bandwidth used by the radar. In this case, the radar may need to put spectral notches in the waveforms it transmits to avoid interfering with other users. Consequently, the radar data will appear randomly sampled in the frequency domain.

An important aspect of practical and real SAR systems is autofocus, which estimates and corrects phase errors caused by uncompensated motion. Although the problem of incorporating autofocus in the recovery of SAR imagery from sparsely sampled data has been addressed in prior work [2], [3], [9], we have found that these approaches, while theoretically elegant, do not perform well in some cases of interest. In particular, while these approaches seem to work well for random phase errors, with realistic SAR scenes they can perform poorly for low-order polynomial phase errors. Although such errors are simpler than purely random ones, they are common in real SAR data and important to handle. Additionally, many prior works addressing autofocus seem to have only considered either relatively small scenes [2], [3] or implicitly employ unrealistic sampling scenarios [3] in which data are not randomly sampled at the true pulse level but in some postprocessed domain, implying that fully sampled radar data are acquired and post-processed, and only then are samples "thrown away."

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In this article, we will develop a polar-format based sparse recovery approach with an integrated autofocus framework and demonstrate its utility on real data, including full-sized large scenes. Our method integrates traditional efficient autofocus algorithms, such as the phase gradient autofocus algorithm (PGA) [10], into the compressive sensing framework. Furthermore, the method we develop operates at the true radar pulse level and thus is applicable to realistic cases in which the raw data are collected in a sparsely sampled fashion from the start.

II. Overview of Synthetic Aperture Radar Imaging

For those readers not familiar with synthetic aperture radar imaging, in this section, we give a short overview. We refer the reader to [11] and [12] for a more comprehensive exposition as well as additional details regarding aspects of the basic polar format algorithm described below. Synthetic aperture radar imaging is a tomographic imaging modality that uses radar echoes from the ground from multiple aspect angles to form an image. To facilitate collection of data from multiple aspect angles, a synthetic aperture radar necessarily is on a moving platform, generally an aircraft or satellite. In this article, we assume the use of the polar format algorithm for image formation; however, the concepts can be generalized in a straightforward manner to other image-formation methods, such as ω-k or backprojection.

The polar format algorithm is a spotlight algorithm that begins by focusing on a single aim point. Let us choose a coordinate system in which the aim point is located at the origin. Let $\mathbf{R}_{\text{radar}}(s) \in \mathbb{R}^3$ denote the position of the radar relative to the aim point at time s. We assume that the received data have been motion compensated relative to the aim point, i.e., the received data have been matched filtered against a reference response that would be generated by a scatterer at the aim point. Let $D(s,t)$ denote the motion-compensated data collected at aperture time (slow-time or pulse-number) s corresponding to the relative (to the aim point) fast-time delay t. The phase history data $\hat{D}(s,f)$ is the Fourier transform of $D(s,t)$ in the fast-time dimension (t variable), where the frequency $f \in [-B/2, B/2]$ spans the bandwidth of the sampled data. Let $R_{\mathbf{p}}(s) = ||\mathbf{R}_{\text{radar}}(s) - \mathbf{p}|| - ||\mathbf{R}_{\text{radar}}(s)||$ for $\mathbf{p} \in \mathbb{R}^3$. Then, the response from an ideal unit amplitude point scatterer at some location $\mathbf{p}$ will yield a contribution to the phase history data of the form

$$\hat{D}_{\mathbf{p}}(s,f) = \exp\left\{-\frac{4\pi}{c} j(f_c + f) R_{\mathbf{p}}(s)\right\} |\hat{w}(f)|^2 \qquad (1)$$

where $|\hat{w}(f)|^2$ is the baseband waveform power spectral density (PSD) and f_c is the RF center frequency.

For simplicity, we model the scene observed by the radar as a field of point scatterers with constant height. Let $I(x,y)$ denote the complex reflectivity of the scatterer at $(x,y,0)$. The total phase history collected by the radar can be obtained by integrating the point scatterer contribution in (1) over the points $\mathbf{p} = (x,y,0)$ weighted by the reflectivity $I(x,y)$, that is

$$\hat{D}(s,f) = \int\int \hat{D}_{(x,y,0)}(s,f) I(x,y) dx dy. \qquad (2)$$

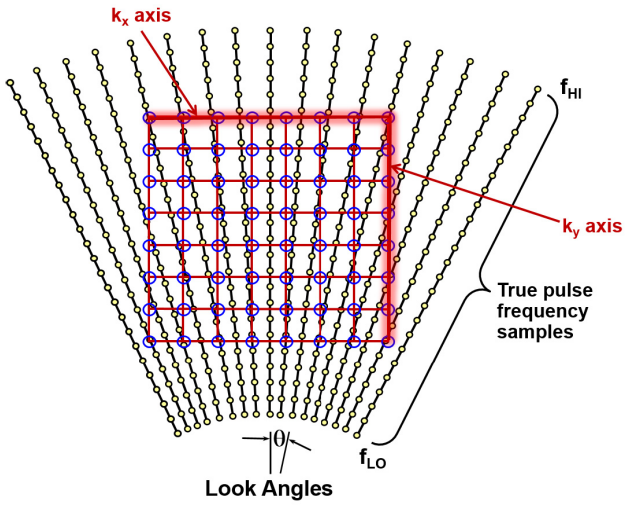


Fig. 1. Illustration of polar format algorithm's polar resampling onto a rectangular grid.

Let $\mathbf{u}(s) = -(\mathbf{R}_{\text{radar}}(s)/||\mathbf{R}_{\text{radar}}(s)||)$ be the unit line-of-sight vector from the radar to the aim point. Let $\mathbf{u}_x$ and $\mathbf{u}_y$ be a basis for the focus target plane, i.e., the plane in $\mathbb{R}^3$ in which scatterers will appear focused in our output image. Let us define the following mapping between image spatial frequency (k_x, k_y) coordinates and the raw slow-time/frequency (s,f) parameterization of the radar data

$$k_y(s,f) \triangleq \frac{2(f+f_c)}{c} \mathbf{u}_y^T \mathbf{u}(s), \quad k_x(s,f) \triangleq \frac{2(f+f_c)}{c} \mathbf{u}_x^T \mathbf{u}(s). \qquad (3)$$

We see from (3) that the RF frequency samples of a given pulse are associated with a radial line in the spatial frequency domain, and thus the raw data samples can be viewed as sampling the spatial frequency domain on a polar grid. Under the assumption of planar wavefronts (which is valid at modest image sizes and long ranges)

$$C(k_x, k_y) \triangleq \hat{D}(s(k_x,k_y), f(k_x,k_y)) \qquad (4)$$

is the Fourier domain representation of an image $I(x,y)$ of the scene observed by the radar, where (3) lets us view (s,f) as functions of (k_x,k_y). We can recover an image of the scene from our collected data via an inverse Fourier transform

$$I(x,y) = \iint \exp(2\pi j(xk_x + yk_y)) \quad C(k_x,k_y) dk_x dk_y. \qquad (5)$$

In practice, the polar format algorithm recovers the image $I(x,y)$ by interpolating the raw phase history data $\hat{D}(s,f)$ onto a rectangular grid in the spatial frequency domain (k_x, k_y space) using the relationship in (3) and is illustrated in Fig. 1. The final image is recovered via a 2-D inverse fast Fourier transform (FFT) operation. Generally, prior to the inverse FFT operation, apodization is applied to $C(k_x,k_y)$ to suppress sidelobes of the system impulse response (IPR).

For certain geometries, the planar wavefront approximation made by the polar format algorithm can restrict the size of the scene over which a focused image can be formed [11]. However, this limitation can largely be mitigated by spatially

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variant filtering to compensate for this defocus in a deterministic fashion [13]–[15]. We discuss this further in Section VI.

III. Proximal Methods for Basis Pursuit Problems

Now, we briefly review the basis pursuit optimization problem we apply to SAR in Section IV and one approach for solving it. First, consider a general problem of the form

$$\text{argmin}_{\boldsymbol{\alpha}} \, f(\boldsymbol{\alpha}) + g(\boldsymbol{\alpha}) \tag{6}$$

where f and g are convex, and f is assumed to be smooth while g may not be.

Suppose we take a gradient step to minimize f, that is

$$\boldsymbol{\alpha}_{new} = \boldsymbol{\alpha} - \delta \nabla f. \tag{7}$$

To attempt to minimize the total objective function, we subsequently try to minimize g while staying close (relative to our stepsize) to $\boldsymbol{\alpha}_{new}$, i.e., we solve

$$\Psi_{g,\delta}(\boldsymbol{\alpha}_{new}) = \text{argmin}_{\boldsymbol{\beta}} \frac{1}{2\delta} ||\boldsymbol{\beta} - \boldsymbol{\alpha}_{new}||_2^2 + g(\boldsymbol{\beta}). \tag{8}$$

Ψ is known as the proximal operator. We restrict to the standard compressive sensing case where

$$f(\boldsymbol{\alpha}) = ||\mathbf{A}\boldsymbol{\alpha} - \boldsymbol{\xi}||^2 \tag{9}$$

is the least-squares residual for a linear operator $\mathbf{A}$ and $g(\alpha) = \lambda ||\boldsymbol{\alpha}||_1$. This choice is frequently referred to as basis pursuit denoising [16]. In the case of $g(\boldsymbol{\alpha}) = \lambda ||\boldsymbol{\alpha}||_1$, where g is parameterized by λ, we simply write $\Psi_{\lambda,\delta}$ and we have the explicit formula

$$\Psi_{\lambda,\delta}(\boldsymbol{\alpha}) = \text{sign}(\boldsymbol{\alpha}) \max(|\boldsymbol{\alpha}| - \delta\lambda, 0) \tag{10}$$

where the expression is interpreted componentwise in $\boldsymbol{\alpha}$ and in the complex case sign is interpreted as the phase. This special case is known as a soft thresholding operator. For details of the derivation of (10) see [1]. The combined step of minimizing f followed by minimizing g (with a quadratic penalty) is

$$\boldsymbol{\alpha}_{k+1} = \Psi_{\lambda,\delta}(\boldsymbol{\alpha}_k - \delta \nabla f(\boldsymbol{\alpha}_k)). \tag{11}$$

Under mild conditions[1] concerning the step-size δ and f, iterating (11) converges to the solution of (6) [17] in the sense that if we let $F(\boldsymbol{\alpha}) = f(\boldsymbol{\alpha}) + g(\boldsymbol{\alpha})$ and $\boldsymbol{\alpha}^*$ is the solution to (6) then

$$F(\boldsymbol{\alpha}_k) - F(\boldsymbol{\alpha}_*) \sim \mathcal{O}\left(\frac{1}{k}\right). \tag{12}$$

This algorithm is frequently referred to as the iterative soft thresholding algorithm (ISTA). Note that convergence here is sublinear.

Beck and Teboulle [17] provided a modification of this algorithm incorporating a momentum term to improve the convergence results

$$\boldsymbol{\Lambda}_{k+1} = \boldsymbol{\alpha}_k + \left(\frac{t_k - 1}{t_{k+1}}\right)(\boldsymbol{\alpha}_k - \boldsymbol{\alpha}_{k-1}) \tag{13}$$

$$\boldsymbol{\alpha}_{k+1} = \Psi_{\lambda,\delta}(\boldsymbol{\Lambda}_k - \delta \nabla f(\boldsymbol{\Lambda}_k)) \tag{14}$$

[1]Specifically that f has Lipschitz continuous gradients and $\delta < \frac{1}{L(f)}$, where $L(f)$ is a Lipschitz constant for ∇f.

where

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \quad t_1 = 1. \tag{15}$$

While the convergence is still sublinear, the overall coefficient is improved as follows:

$$F(\boldsymbol{\alpha}_k) - F(\boldsymbol{\alpha}_*) \sim \mathcal{O}\left(\frac{1}{k^2}\right). \tag{16}$$

This algorithm is referred to as fast iterative soft thresholding algorithm (FISTA).

IV. Polar Format-Based Sparse Reconstruction

Now, we develop the sparse reconstruction framework. As mentioned, we formulate the algorithm in the context of SAR imaging using the polar format algorithm. Let $\mathbf{x} \in \mathbb{C}^{MN}$ denote the fully sampled (i.e., uniformly sampled in slow-time and frequency) radar complex phase history data. Here, N is the number of radar pulses and M is the number of frequency samples collected. Note that we switch between viewing elements of $\mathbb{C}^{MN}$ as MN dimensional vectors and $M \times N$ arrays, depending on context. Explicitly, we associate an $M \times N$ array with an MN vector by stacking consecutive columns of the array.

Recall that the first stage of the polar format algorithm (after motion compensation and pulse compression or dechirping the data) is to interpolate data from the polar grid on which it is naturally sampled onto a rectangular grid in the spatial frequency domain (see Fig. 1). The interpolation operation from a polar to rectangular grid can be viewed as a linear transformation of $\mathbf{x}$. Let $\mathbf{y} \in \mathbb{C}^{mn}$ denote spatial frequency domain samples obtained by interpolation of the raw phase history $\mathbf{x}$ onto an $m \times n$ rectangular grid (corresponding to an $m \times n$ pixel image). Each interpolated sample of $\mathbf{y}$ is a linear combination of the raw samples in $\mathbf{x}$. Thus, there is $\mathbf{C} \in L(MN, mn)$ (where $L(MN, mn)$ denotes the space of linear transformations from $\mathbb{C}^{MN}$ to $\mathbb{C}^{mn}$) corresponding to the interpolation operator such that

$$\mathbf{y} = \mathbf{C}\mathbf{x}. \tag{17}$$

If we let $\mathbf{F}$ denote the inverse 2-D FFT operation (assumed to be normalized so it is unitary), then the image $\mathbf{z} \in \mathbb{C}^{mn}$ with m range samples and n cross-range samples is related to the phase history data via

$$\mathbf{z} = \mathbf{F}\mathbf{C}\mathbf{x}. \tag{18}$$

We will define an operator $\mathbf{P}$ implicitly by first defining its adjoint $\mathbf{P}^H$. We will let $\mathbf{P}^H = \mathbf{F}\mathbf{C}$ denote the "polar format" operator. Note that $\mathbf{P} = \mathbf{C}^T\mathbf{F}^H$ is then an operator that takes an image and applies a 2-D FFT (the FFT corresponding to $\mathbf{F}^H$ assuming our Fourier transforms are unitary) and then does the reverse set of interpolations from rectangular samples back to the raw pulse–frequency domain (where the reverse interpolations correspond to $\mathbf{C}^T$). Note that we write $\mathbf{C}^T$ rather than $\mathbf{C}^H$ because for basebanded data the interpolation filters would be real filters (applied to complex data). We colloquially refer to the action of the operator $\mathbf{P}$ as "inverse" polar formatting.

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In the previous paragraph, we described the action of $\mathbf{C}^T$ as reversing the 2-D interpolations performed by $\mathbf{C}$ during the polar format operation. Let us now elaborate on that characterization. Since $\mathbf{C}$ interpolates from the samples on a polar grid to samples on a rectangular grid, the adjoint operator $\mathbf{C}^T$ provides a map from the samples on the rectangular grid back to samples on the polar grid. Furthermore, under $\mathbf{C}^T$ the contribution of each rectangular grid sample to a given polar grid sample is equal to how much that polar grid sample contributed to the rectangular grid sample under $\mathbf{C}$. Additionally, each row of $\mathbf{C}$ is relatively sparse because for the polar format resampling we typically use bandlimited interpolation filters that have the form of weighted truncated sinc functions [12], and these filters decay relatively rapidly. From this, we see that each polar grid sample of $\mathbf{C}^T\mathbf{y}$ will be formed as the weighted sum of a small number of nearby rectangular grid samples from $\mathbf{y}$ (nearby in the spatial frequency domain). As long as our sampling in both the polar and rectangular grids is fine enough that our data remain bandlimited in both domains, we can think about $\mathbf{C}^T$ as defining a set of interpolation filters from the rectangular grid samples back to the polar grid samples. Rather than computing the exact interpolation filter coefficients that would correspond to $\mathbf{C}^T$, we simply use another set of bandlimited interpolation filters that interpolate from the rectangular grid samples back to the polar sampling of the raw data. This approximation of $\mathbf{C}^T$ seems to work well in practice. Note that we do not ever explicitly compute the matrices $\mathbf{C}$ or $\mathbf{C}^T$, and only compute their actions; for both $\mathbf{C}$ and $\mathbf{C}^T$ the overall 2-D interpolation is realized as successive 1-D interpolations [12]. The use of approximate adjoint operators in iterative image reconstruction algorithms is common, see [18] for further discussion.

Let us now turn to formulating the sparse reconstruction problem. Although the raw radar data $\mathbf{x}$ is nominally uniformly sampled, we assume that we actually collect $K \leq MN$ samples and there is a sampling matrix $\mathbf{B} \in L(MN, K)$ such that the samples we actually have access to are $\mathbf{w} \in \mathbb{C}^K$

$$\mathbf{w} = \mathbf{B}\mathbf{x}. \tag{19}$$

The problem we wish to solve is

$$\text{argmin}_{\mathbf{z}} \, ||\mathbf{B}\mathbf{P}\mathbf{z} - \mathbf{w}||_2^2 + \lambda||\mathbf{z}||_1. \tag{20}$$

This fits into the framework of Section III with

$$f(\mathbf{z}) = ||\mathbf{B}\mathbf{P}\mathbf{z} - \mathbf{w}||_2^2 \tag{21}$$

and

$$g(\mathbf{z}) = \lambda||\mathbf{z}||_1. \tag{22}$$

In the case that the entries of $\mathbf{B}$ are only 1s or 0s, it simply masks samples we do not collect, we can interpret $\mathbf{P}\mathbf{z}$ as inverse polar formatting an image $\mathbf{z}$ back to raw phase history data so $f(\mathbf{z})$ is the least-squares error between the actual phase history samples we collect and the corresponding samples from the "simulated" phase history data obtained by inverse polar formatting $\mathbf{z}$.

The relevant gradient (written using the Wirtinger calculus [19], [20]) is

$$\nabla_{\mathbf{z}^*} f = \mathbf{P}^H\mathbf{B}^H(\mathbf{B}\mathbf{P}\mathbf{z} - \mathbf{w}). \tag{23}$$

The overall ISTA iteration is thus

$$\mathbf{z}_{k+1} = \Psi_{\lambda,\delta}(\mathbf{z}_k - \delta\mathbf{P}^H\mathbf{B}^H(\mathbf{B}\mathbf{P}\mathbf{z}_k - \mathbf{w})). \tag{24}$$

The full algorithm using the FISTA approach is shown in Algorithm 1.

V. Autofocus

In general, SAR imagery formed solely based on position/velocity information from the radar GPS/INS system will have some amount of defocus due to motion compensation errors. Such errors can be modeled as the data being corrupted by unknown phase errors on a pulse-by-pulse basis [10]–[12]. These errors can be incorporated into the sparse reconstruction process by positing a diagonal operator $\mathbf{D}(\boldsymbol{\phi}) \in L(MN, MN)$ parameterized by the vector $\boldsymbol{\phi} \in \mathbb{R}^N$, which applies these phase errors to the phase history data. Each component of $\boldsymbol{\phi}$ corresponds to the phase error for each pulse and we write $\boldsymbol{\phi}[l]$ for the lth component of $\boldsymbol{\phi}$. We can write $\mathbf{D}(\boldsymbol{\phi})$ explicitly as the $MN \times MN$ matrix

$$\mathbf{D}(\boldsymbol{\phi}) = \begin{pmatrix} e^{j\phi[1]}\mathbf{I}_M & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & e^{j\phi[2]}\mathbf{I}_M & \mathbf{0} & \vdots \\ \vdots & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & e^{j\phi[N]}\mathbf{I}_M \end{pmatrix} \tag{25}$$

where the lth M-dimensional identity matrix along the diagonal corresponds to the M frequency samples collected on the lth pulse. We can extend the sparse recovery problem to the joint estimation of the unknown phase errors $\boldsymbol{\phi}$ and focused image $\mathbf{z}$, i.e., we solve

$$\text{argmin}_{\mathbf{z},\boldsymbol{\phi}} \, ||\mathbf{B}\mathbf{D}(\boldsymbol{\phi})\mathbf{P}\mathbf{z} - \mathbf{w}||_2^2 + \lambda||\mathbf{z}||_1, \tag{26}$$

which is the approach taken in [2], [3], and [9]. This approach is referred to as sparsity-driven autofocus (SDA) in [9].

In [2] and [9], it is proposed to solve (26) using an alternating minimization approach. In this case, one alternates between updating the estimate of the image given the current phase error estimate and then updating the phase error given the current image. Let $\mathbf{z}_i$, and $\boldsymbol{\phi}_i$ be the image and phase error estimate on the ith iteration. We update the image by solving (27)

$$\mathbf{z}_{i+1} = \text{argmin}_{\mathbf{z}} \, ||\mathbf{B}\mathbf{D}(\boldsymbol{\phi}_i)\mathbf{P}\mathbf{z} - \mathbf{w}||_2^2 + \lambda||\mathbf{z}||_1. \tag{27}$$

To express the SDA phase update used in [2] and [9], let

$$\tilde{\mathbf{w}} = \mathbf{B}^H\mathbf{w} \tag{28}$$

$$\mathbf{v}_{i+1} = \mathbf{D}(\boldsymbol{\phi}_i)\mathbf{P}\mathbf{z}_{i+1}. \tag{29}$$

Note that $\tilde{\mathbf{w}}, \mathbf{v}_{i+1} \in \mathbb{C}^{MN}$. Let us write $\tilde{\mathbf{w}}[:, l]$ for the M-dimensional vector corresponding to the lth column of $\tilde{\mathbf{w}}$, and similarly for $\mathbf{v}_{i+1}[:, l]$. Then, the SDA update for the estimate of the phase error of the lth pulse can be written

$$\delta\boldsymbol{\phi}_{i+1}[l] = \angle\{\mathbf{v}_{i+1}^H[:, l]\tilde{\mathbf{w}}[:, l]\} \tag{30}$$

$$\boldsymbol{\phi}_{i+1}[l] = \boldsymbol{\phi}_i[l] + \delta\boldsymbol{\phi}_{i+1}[l]. \tag{31}$$

Note that the phase error update equation in (30) has a very simple intuitive interpretation. To compute $\mathbf{v}_{i+1}$, we take our updated image estimate $\mathbf{z}_{i+1}$ and current phase error

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Algorithm 1 FISTA Approach for Sparse Polar Format SAR Reconstruction

1: **procedure** FISTASAR
2: $k \leftarrow 1$
3: $\mathbf{z}_0 \leftarrow$ *Initial image to start optimization. Could be* 0
4: $\Lambda_1 \leftarrow \mathbf{z}_0$
5: $t_1 \leftarrow 1$
6: **while** $\mathbf{z}_k$ not converged **do**
7: $\mathbf{z}_k \leftarrow \Psi_{\lambda,\delta}(\Lambda_k - \delta \mathbf{P}^H \mathbf{B}^H (\mathbf{B}\mathbf{P}\Lambda_k - \mathbf{w}))$ ▷ Ψ is simply soft thresholding, δ chosen by backtracking line search
8:
9: $t_{k+1} = \frac{1+\sqrt{1+4\ t_k^2}}{2}$
10: $\mathbf{\Lambda}_{k+1} \leftarrow \mathbf{z}_k + \left(\frac{t_k - 1}{t_{k+1}}\right)(\mathbf{z}_k - \mathbf{z}_{k-1})$
11: $k \leftarrow k + 1$
12: **go to** 7
13: **return** $\mathbf{z}_k$

estimate $\boldsymbol{\phi}_i$ and "simulate" based on our imaging model, the corresponding phase history data. We then compare the average residual phase difference on each pulse between our simulated phase history data $\mathbf{v}_{i+1}$ and the actual measured phase history data $\mathbf{w}$. This difference is then used to update our total estimate of the phase errors in (31). The SDA algorithm proceeds by alternating between (27) and (31).

At a high level, the alternative autofocus approach we propose operates similar to SDA in that it alternates between updating the image and the phase error estimate; however, our approach replaces the update in (30) with an update strategy that incorporates traditional autofocus algorithms. As in SDA, on each iteration we compute a new image $\mathbf{z}_{i+1}$ by solving (27). However, we then form

$$\tilde{\mathbf{z}}_{i+1} = \mathbf{P}^H \mathbf{P} \mathbf{z}_i. \tag{32}$$

Note that $\tilde{\mathbf{z}}_{i+1}$, which is obtained by essentially simulating phase history data having the scatterers in $\mathbf{z}_{i+1}$ and then polar formatting it, tends to look like conventional SAR imagery, and conventional autofocus can be applied to it without issues.

To update $\boldsymbol{\phi}_i$, we use the phase correction obtained by applying a conventional autofocus algorithm such as PGA to $\tilde{\mathbf{z}}_{i+1}$. To fit into the same update strategy as the original SDA algorithm, we need a phase error estimate at the pulse level. However, conventional autofocus algorithms applied to the polar format image $\tilde{\mathbf{z}}_{i+1}$ will provide a phase correction $\delta\tilde{\boldsymbol{\phi}}(k_x)$ as a function of the image cross-range spatial frequency k_x. To turn this into a phase error estimate at the pulse level, observe that (3) lets us link cross-range spatial frequency k_x with slow-time (pulse) by fixing f equal to the center frequency f_{mid}. Thus, we can view the autofocus solution from conventional autofocus as a function of slow-time (pulse) via

$$\delta\boldsymbol{\phi}(s) = \delta\tilde{\boldsymbol{\phi}}(k_x(s, f_{\text{mid}})) \tag{33}$$

and use this phase error estimate for the current image to update our overall pulse-level phase error estimate as in (31). It is important to note that we are treating slow-time as continuous in (33), but since pulses are discrete in reality, interpolation is implied. Precisely one must associate the grid of samples in the k_x domain on which the polar format image is formed with a discrete set of pulse times via (3). This association allows one to interpolate the autofocus correction from the k_x domain grid back to raw pulse times. This is what is implicitly meant in writing (33). See Fig. 1 for an illustration of the relationship between spatial frequency domain coordinates (k_x, k_y) and raw pulses.

Thus, the overall algorithm becomes as shown in Algorithm 2 where in line 11, *Auto* refers to any conventional autofocus algorithm, e.g., PGA, mapDrift, or entropy minimization. We use Algorithm 1 to solve the problem on line 7 but any preferred solver could be used, if desired. Also, since we iterate and update the autofocus after each solution, the problem on line 7 need not be solved to high accuracy on each iteration. Let us also mention that the reason that we do not apply conventional autofocus directly to $\mathbf{z}_i$ is that the sidelobe structure of $\mathbf{z}_i$ may, in general, be rather different from conventional SAR imagery as many pixels may be thresholded to 0 for large values of λ, which traditional autofocus algorithms may not be tuned for.[2]

From the above description, our approach can be viewed as operating within the framework of (26) but using conventional autofocus to accelerate the convergence of alternating minimization to a focused image. Our approach to sparse recovery with integrated autofocus is summarized in Algorithm 2.

VI. Wavefront Curvature Correction

This section briefly addresses methods to correct for limitations arising from the use of the polar format imaging operator in the reconstruction process. On a first read-through, this section may be skipped by readers who are not experts in SAR imaging without hindering understanding of the results presented in the subsequent sections.

As mentioned in Section II, the polar format algorithm can in some cases be limited by the planar wavefront approximation under which it is derived. The implication of this

[2]However, we have run the algorithm with autofocus applied directly to $\mathbf{z}_i$ and obtained good results but obtained slightly better results when applied to $\tilde{\mathbf{z}}_i$. It is possible that different autofocus implementations may work better in one way or the other.

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Algorithm 2 Polar Format Sparse Reconstruction With Autofocus

procedure POLARSPARSEAUTO
  $k \leftarrow 0$
  $\mathbf{z}_k \leftarrow 0$
  $\boldsymbol{\phi}_k \leftarrow 0$
  $\hat{\mathbf{z}}_k \leftarrow 0$
  while $\mathbf{z}_k$ not converged **do**
    $\mathbf{z}_{k+1} \leftarrow \text{argmin}_{\mathbf{z}} ||\mathbf{BD}(\boldsymbol{\phi}_k)\mathbf{Pz} - \mathbf{w}||_2^2 + \lambda||\mathbf{z}||_1$ ▷ Can be solved using Algorithm 1 (with $\mathbf{B}$ redefined as $\mathbf{BD}(\boldsymbol{\phi}_k)$). Use $\hat{\mathbf{z}}_k$ as the starting point for the optimization.
    $\tilde{\mathbf{z}}_{k+1} \leftarrow \mathbf{P}^H\mathbf{P}\mathbf{z}_{k+1}$
    $\delta\boldsymbol{\phi}_{k+1}(k_x),\ \hat{\mathbf{z}}_{k+1} \leftarrow Auto(\tilde{\mathbf{z}}_{k+1})$ ▷ *Auto* is any conventional autofocus algorithm. It produces a phase correction $\delta\boldsymbol{\phi}$ and focused image $\hat{\mathbf{z}}$ used to initialize the next optimization step.
    $\boldsymbol{\phi}_{k+1}(s) = \boldsymbol{\phi}_k(s) + \delta\boldsymbol{\phi}_{k+1}(k_x(s, f_{mid}))$
    $k \leftarrow k + 1$
    go to 7
  return $\mathbf{z}_k$

limitation is that the scene size over which an image can be formed without defocus may be relatively small, depending on the geometry. One approach to avoid these limitations is to abandon the polar format algorithm and use an alternate image formation framework such as backprojection (or fast variants) or $\omega - k$ as the general framework of Section IV easily accommodates other image formation approaches by appropriately redefining $\mathbf{P}$ and $\mathbf{P}^H$.

An alternate approach that we take here is to exploit the fact that the limitations of the basic polar format algorithm can be largely mitigated by spatially variant postprocessing [13]–[15]. These approaches partition the initial polar format image into chips and apply a local 2-D phase correction to the chips in the frequency domain to improve the focus of each chip and enable the image to retain its focus over very large scenes. Furthermore, in addition to a phase correction, an inexpensive spatially variant polar resampling operation can be done to extend the feasible scene size arbitrarily large [15]. With this simple modification, the polar format algorithm is one of the fastest image formation algorithms (although it does have the same general computational complexity as fast-backprojection) while having minimal limitations on achievable scene size. Speed is important for the feasibility of sparse recovery on full-scale SAR data because each iteration of ISTA or FISTA requires executing the image formation algorithm (and its inverse/transpose).

Incorporating wavefront curvature corrections into the formulation of Section IV amounts to modifying our definition of $\mathbf{P}^H$ to be $\mathbf{P}^H = \mathbf{SFC}$, where $\mathbf{S}$ denotes any spatially variant post-filtering. This implicitly redefines $\mathbf{P}$ to be $\mathbf{P} = \mathbf{C}^T\mathbf{F}^H\mathbf{S}^H$. Using the same line of reasoning developed for the adjoint operator $\mathbf{C}^T$ in Section IV, rather attempting to compute the adjoint $\mathbf{S}^H$ explicitly or exactly, we implement $\mathbf{S}^H$ as an operator that approximately inverts the spatially variant filtering done by $\mathbf{S}$. Our results in Section VIII implement this more general version of $\mathbf{P}$ and $\mathbf{P}^H$.

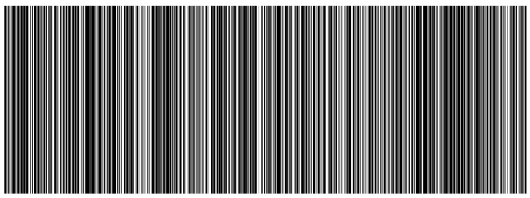
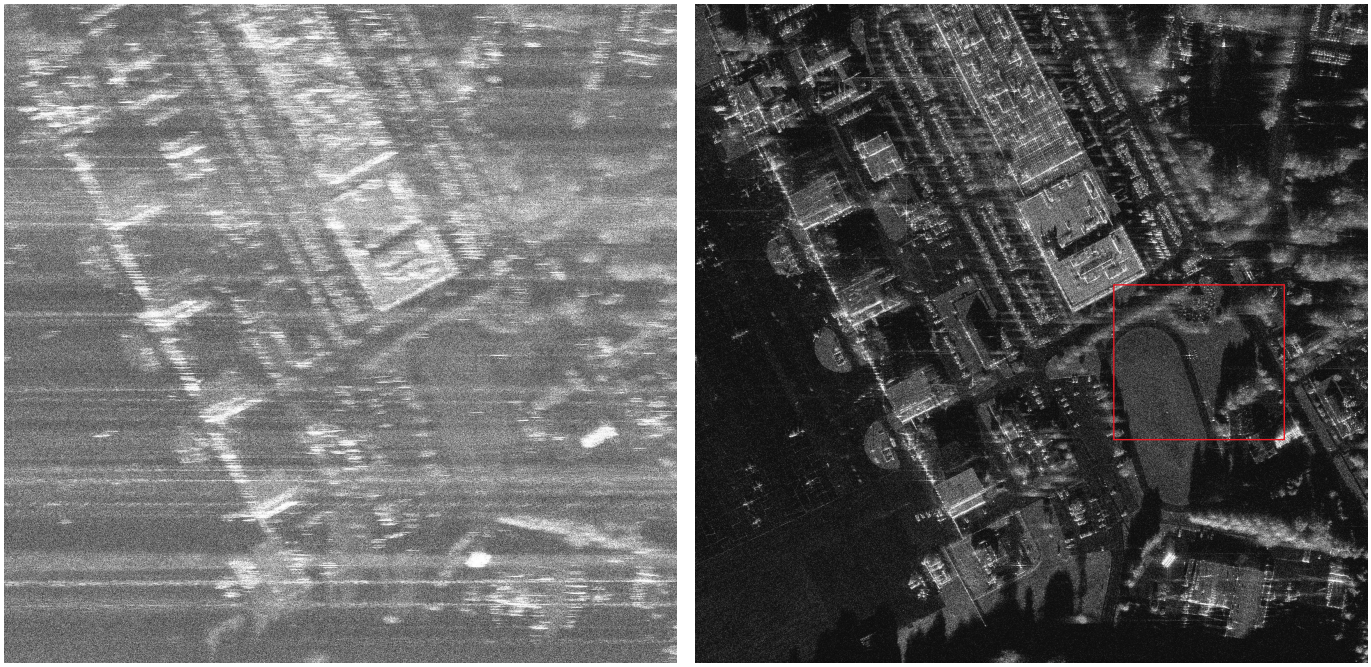


Fig. 2. Slow-time pulse mask for random sampling; 39% of pulses in original data set are kept.

VII. PRIOR WORK IN THE LITERATURE

Before presenting some example results using our algorithm, we would like to highlight some differences between the results we present here and prior published results. The first difference is that many prior results such as [1] appear to assume a random sampling of the data from a rectangular grid in the image spatial frequency domain, i.e., sampling after conventional image formation has already been applied. Randomly sampling in this way simplifies the problem substantially because the operator linking the image domain to the data domain becomes a simple 2-D FFT. In [1], the unitary property of the FFT is exploited to simplify the sparse recovery problem. Unfortunately, this rectangular grid sampling strategy is not, in general, physically realizable, because the pulses that the radar collects actually correspond to a polar sampling of the spatial frequency domain as discussed in Section II (see Fig. 1). We note that this conflation of the raw samples of pulse/frequency data with data interpolated to a rectangular grid in the spatial frequency domain may be acceptable in some cases; however, in general, the distinction is important. In contrast, the operator we use to relate the image domain to the raw data is the full image formation operator $\mathbf{P}$. We should mention that [2] does randomly sample at a true pulse level and uses a full fast backprojection/reprojection approach to relate the image domain to the data domain.

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(a) (b)

Fig. 3. (a) Image using randomly sampled data (39% of pulses) with quadratic phase error and conventional polar format imaging. For comparison, (b) shows an image with conventional polar format imaging applied to fully sampled data with no phase errors. Region shown in Fig. 5 corresponds to red box in (b).

The second limitation concerns autofocus. For large low-order polynomial phase errors with realistic full-size scenes, we were unable to obtain good results applying the alternating minimization scheme used by SDA to solve (26) which is the approach used in [2], [3], and [9]. The results in [3] and [9], which utilize SDA, appear to be run on small scenes with relatively few scatterers. For example, the results in [3] use small 128×128 chips with a small number of scatterers and no cultural clutter background. In these cases, it seems that the regularization term can be set strong enough to focus the isolated point scatterers and enable quick convergence of the SDA alternating minimization scheme. However, real large SAR scenes are more complicated, and it is important to not over-regularize the optimization problem and suppress distributed cultural clutter. In our experience, in order for SDA to achieve good focusing, it is necessary to use a relatively large ℓ_1 regularization term, which for real SAR imagery has the effect of suppressing the SAR background to an unacceptable level. We illustrate this in Section IX. We note that [3] uses ℓ_p norms with $p < 1$; however (on the basis of admittedly limited experimentation with the $p < 1$ case), we did not find that this eliminated the need to over-regularize the problem. These issues, which we demonstrate in Sections VIII and IX, motivate our incorporation of conventional autofocus in the sparse reconstruction process.

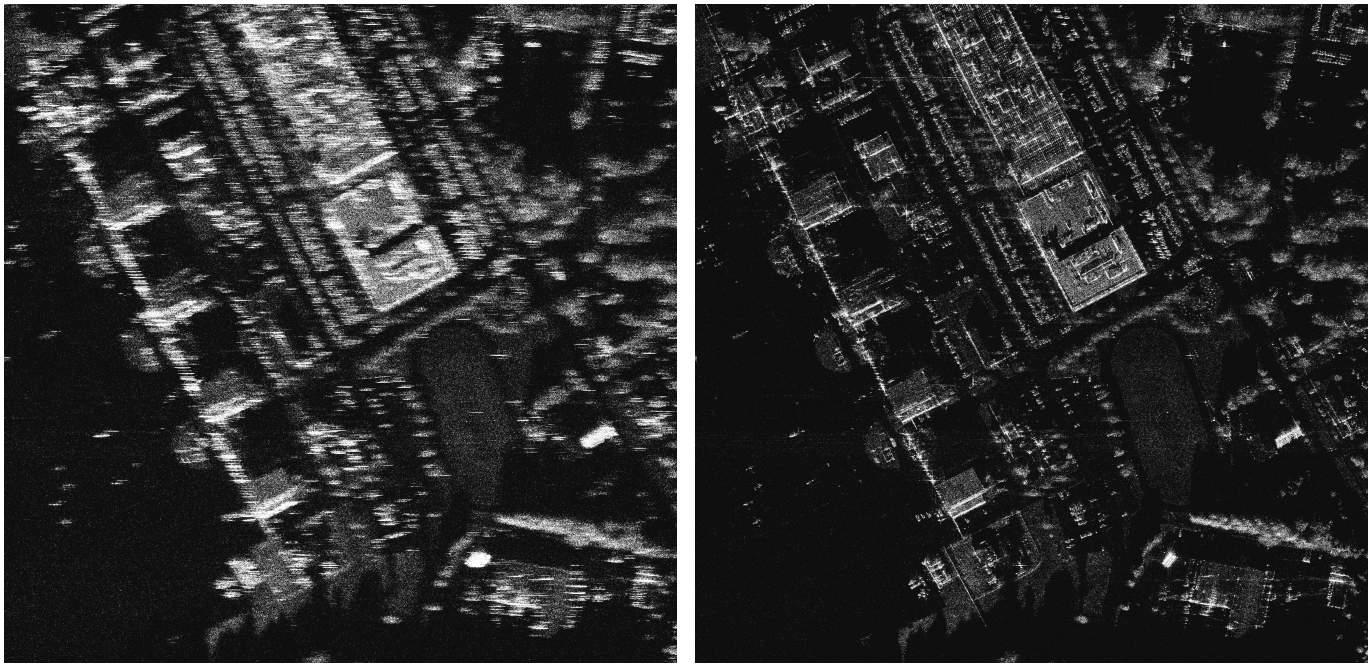
VIII. Example Results

We use data from an experimental X-band SAR with a 700-MHz bandwidth linear frequency modulation (LFM) waveform for these tests. We reconstruct a full image of 8192 $\times$ 16384 pixels with 0.25 sample spacing. For interpretability, what is shown here are smaller chips of that image. We use FISTA for the sparse reconstruction with backtracking search, and for the SDA example with alternating minimization, we alternate updates to the sparse solution (in our case iterations of FISTA) with updates of the phase correction as in [2] and [3]. The raw phase history data are 44 000 pulses $\times$ 32 000 frequency samples. Note that this size is nearly two orders of magnitude larger than any similar results in the literature for sparse SAR imaging that we are aware of (the largest being the 5000×3500 sample case in [1], which operates in the image spatial frequency domain and not in the raw data domain). The original data were collected with a constant PRF, and we randomly sample the pulses so that we retain 39% of the pulses. The pulse mask used is shown in Fig. 2. We intentionally add a large quadratic phase error to the data to simulate a common class of motion-compensation errors. In Fig. 3(a), we show the results of conventional polar format imaging with partial data, which, of course, has very high cross-range sidelobes because of the random sampling.

In Figs. 4(a) and 5(a), we show the results of using alternating minimization with the sparsity-driven autofocus formulation of (26). Note that the sidelobes caused by the random sampling have been mitigated but the scatterers still have substantial broadening of the impulse response.

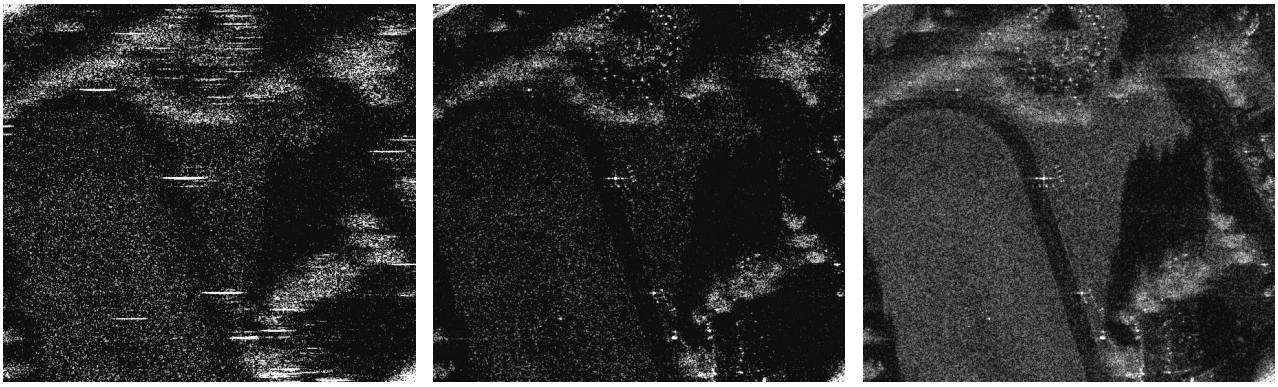
In Figs. 4(b) and 5(b), we apply the method of Algorithm 2. Although some of the distributed ground clutter in the scene is suppressed as expected, overall the image is well focused and interpretable. For comparison, in Figs. 3(b) and 5(c), we show imagery formed using the polar format algorithm applied to fully sampled data with no phase errors.

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(a) (b)

Fig. 4. Sparse reconstruction methods applied to randomly sampled data (39% of pulses) with quadratic phase error. (a) Reconstruction using SDA algorithm from [9]. (b) Reconstruction with proposed Algorithm 2.



(a) (b) (c)

Fig. 5. Zoomed-in view images from sparse reconstruction methods and conventional imaging. All images correspond to the region indicated by the red box in Fig. 3(b). (a) Reconstruction using SDA, corresponding to Fig. 4(a). (b) Reconstruction with proposed algorithm, corresponding to Fig. 4(b). (c) Conventional imaging, fully sampled, and no phase error, corresponding to Fig. 3(b).

Note that as described in Section V, both SDA and our Algorithm 2 must solve the same image update equation shown in (27) on each iteration. The only difference is the approach for updating the phase error, which even in the case of Algorithm 2 that uses conventional autofocus, is still an order of magnitude less expensive than the image update. Thus, the execution time for the image in Fig. 4(b), while a bit longer, was of a similar order of magnitude as for Fig. 4(a). In Table I, we summarize the reconstruction time and the image focus measured using image entropy [21], [22]. Recall that smaller image entropy generally corresponds to more well-focused images.

TABLE I

SUMMARY METRICS FOR FIG. 4. LOWER IMAGE ENTROPY CORRESPONDS TO BETTER FOCUSED IMAGERY

Method	Computation Time	Entropy
Algorithm 2	51.6 minutes	10.96
SDA	41.3 minutes	11.99

IX. PARAMETRIC AUTOFOCUS ANALYSIS

In this section, we illustrate the performance of the proposed method somewhat more quantitatively. As before, we use data from an experimental X-band SAR with a 700-MHz bandwidth LFM waveform for these tests. To facilitate a parametric sweep, we preprocess the data by decimating in range and pulse. The resulting phase history data are 3200 pulses $\times$ 4400 frequency samples which correspond to a smaller image covering only 600 $\times$ 2400 pixels at 0.25 m resolution. Although we still use the full polar format operator for image formation, the decimated data set can be processed

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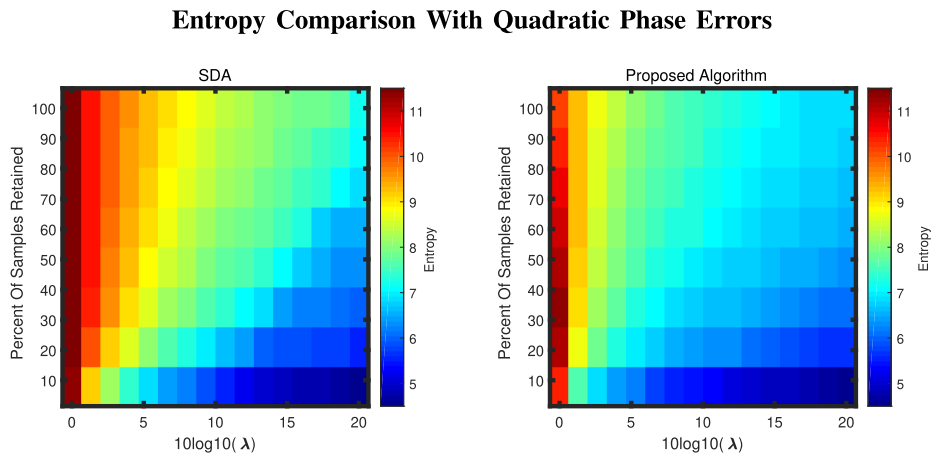


Fig. 6. Image entropy for SDA and proposed algorithm as regularization parameter λ and percentage of samples retained in the 2-D phase history domain is varied. Note that we have normalized lambda so that the smallest value used in our studies corresponds to a value of 1. Phase error is quadratic for these cases. Lower entropy corresponds to better focus.

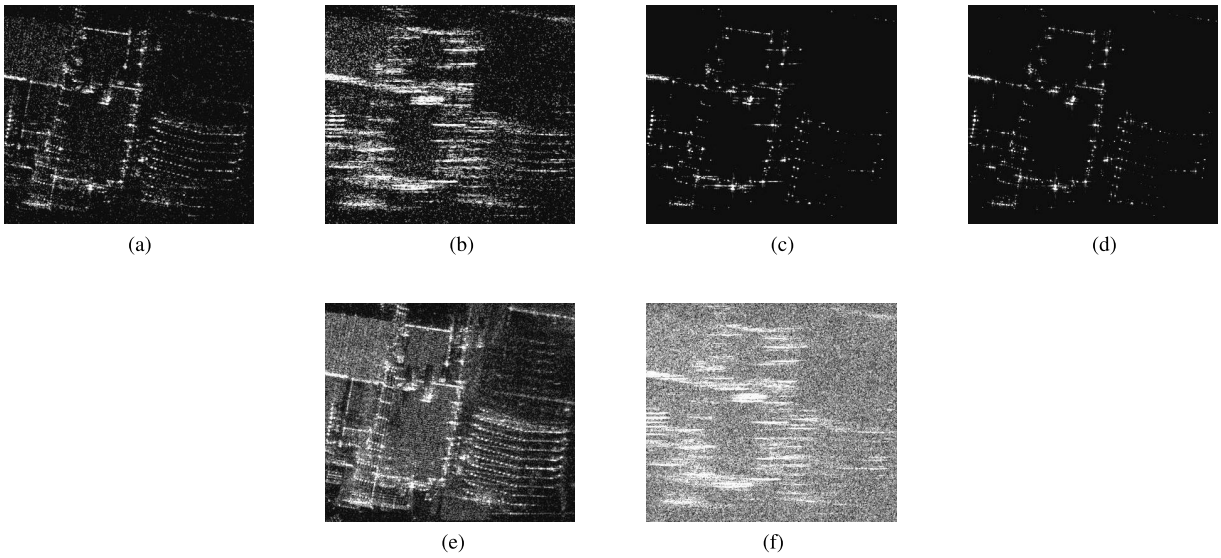


Fig. 7. (a)–(d) Reconstructions using SDA and the proposed algorithm with data randomly sampled in the 2-D phase history domain such that 22% of phase history samples are retained. A quadratic phase error is introduced into the data prior to reconstruction. In (a) and (b), the normalized regularization parameter is set to $10\log_{10}(\lambda) = 11.5$ (near the middle of that shown on the horizontal axis in Fig. 6). In (c) and (d), the normalized regularization parameter $10\log_{10}(\lambda) = 18.2$ is near the top of that shown in Fig. 6. (a) Proposed algorithm with medium regularization, 22% of samples kept, with quadratic phase error. (b) SDA with medium regularization, 22% of samples retained, with quadratic phase error. (c) Proposed algorithm with large regularization, 22% of samples retained, with quadratic phase error. (d) SDA with large regularization, 22% of samples retained, with quadratic phase error. (e) Fully sampled image, no phase error. (f) Conventional polar format imaging with 22% of samples retained and quadratic phase error.

in about 130 s for our proposed algorithm and 179 s using SDA. For these experiments, we will consider a different random sampling strategy of the data. Here, we look at a case in which the phase history samples available are obtained by randomly sampling jointly in the 2-D pulse/frequency domain.

In Fig. 6, we show image entropy as the regularization parameter λ is varied over two orders of magnitude, and the percentage of samples retained is varied from 8% to 100 %. Note that we have normalized lambda so that the smallest value used in our studies corresponds to a value of 1, and the largest value corresponds to 100. From Fig. 6, we can clearly see that Algorithm 2 is able to achieve lower entropy (corresponding to better-focused imagery) than SDA, especially for smaller values of the regularization parameter. This result illustrates the ability of our proposed method to achieve better-focused imagery over a larger domain of the regularization parameter λ.

In Fig. 7, we show the resulting images for medium and large regularization parameters when 22% of the samples are retained. Note that with the proposed algorithm, we can obtain focused imagery that has mitigated the cross-range sidelobes that are due to random sampling for much smaller values of λ than is possible with SDA. We generally want to choose λ only as large as is necessary to eliminate image artifacts due to random sampling. Keeping λ as small as possible while

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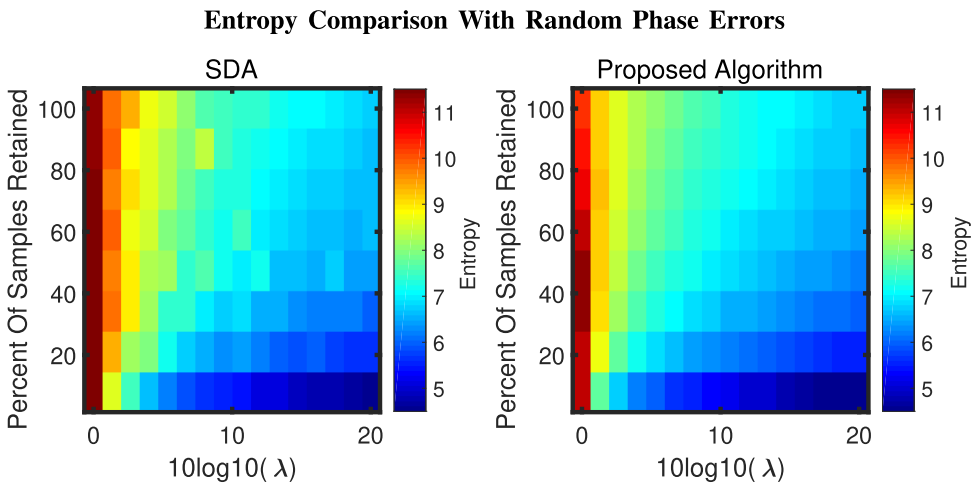


Fig. 8. Image entropy for SDA and proposed algorithm as regularization parameter λ and percentage of pulses retained is varied. Note that we have normalized λ so that the smallest value used in our studies corresponds to a value of 1. Phase errors are random (obtained by applying moving average filter to draws of Gaussian distribution on each pulse). Lower entropy corresponds to better focus.

still eliminating artifacts from random sampling ensures that we do not suppress important subtle features of the image. For small-to-medium values of λ, SDA simply reconstructs the defocused scatterers and image focus is not improved. For large values of λ, SDA is able to eliminate the cross-range defocus from the quadratic phase, but the cost is that many weaker scatterers are suppressed, reducing image interpretability. It should be noted that for large values of λ where the image becomes extremely sparse, SDA may in fact slightly outperform Algorithm 2 which uses conventional autofocus, as conventional autofocus algorithms are generally not tuned for SAR scenes that look like this. Fig. 7(c) and (d) illustrates the region of the parameter space where SDA outperforms conventional methods. However, the image quality for smaller λ, where Algorithm 2 outperforms SDA, is in our opinion much preferred. We also mention that while this 2-D random sampling strategy is different from the 1-D random pulse sampling used in Section VIII, the trend seen in Fig. 6 also is reproduced for the case of 1-D random sampling in the pulse domain.

We have observed that for some types of phase errors, SDA performs similar to our proposed method. In particular, for noise-like phase errors that are largely independent of pulse to pulse, SDA performs quite well even for small values of the regularization parameter. Fig. 8 shows entropy versus samples retained and the regularization parameter for the same data as in Fig. 7 but with phase errors of a random nature. More precisely, the phase error is obtained by applying a moving average filter to phase errors drawn from a Gaussian distribution on each pulse. Note that while our method appears to perform slightly better, the differences are not as dramatic for small regularization values as for the quadratic phase error in Fig. 6. Of course, since phase errors encountered in real data are generally a composite of low-order polynomial phase errors with high-order terms superimposed, it is important to be able to handle both cases for realistic scenes and data sizes, which our method does.

X. Conclusion

We have developed a polar format-based algorithm for recovering SAR imagery from sparsely sampled data in the presence of motion compensation/phase errors. Our approach incorporates conventional SAR autofocus algorithms into the sparse recovery problem. We have demonstrated the effectiveness of this approach on real full-scale SAR imagery and provided quantitative comparisons with existing methods.

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